

HE2003 Econometrics I

Tutorial 1 (Week 2) – Problems & Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

September 28, 2026

Overview

- **Lecture coverage:** Lecture 1 — *Introduction: why we study econometrics, correlation vs. causality, the linear regression model previewed via the California class-size example, Type I/Type II errors, and randomised controlled trials.*
- **Questions:** 4 questions from Tutorial Question Set No. 1; solutions extend the official Tut 1 slides (12 pages, Aug 24, 2026).
- **Notation:** as in lectures — β_0, β_1 population parameters; u_i population error term (never $\hat{u}_i$ at this stage, since no data has been fitted yet); $E[\cdot]$ population expectation; $E[\cdot | \cdot]$ conditional expectation.

Question 1 (Correlation vs. causality: the three cases)

Problem

Simply observing a correlation between two variables X and Y is not enough to conclude that X has a causal effect on Y . When X and Y are correlated, there are several possible cases. Explain these cases with specific examples. (*Tutorial Question Set No. 1, Q1.*)

Solution

Method 1: Enumerate the three structural cases (official route)

Step 1: Why correlation alone is not enough. A sample correlation $\text{Corr}(X, Y) \neq 0$ tells us only that X and Y move together on average; it carries no information about the *direction* of any causal arrow, or about whether an arrow exists at all. Formally, if we naively estimate $Y_i = \beta_0 + \beta_1 X_i + u_i$ and find $\hat{\beta}_1 \neq 0$, this is consistent with several very different underlying data-generating processes. There are three mutually exclusive structural explanations for an observed correlation between X and Y .

Case (i): X causes Y (forward causality). $X \rightarrow Y$: changes in X genuinely bring about changes in Y , and there is no feedback from Y to X and no third variable driving both.

Official example (windmills and wind). The faster windmills are observed to rotate, the stronger the wind is observed to be. Naively one might ask: does the wind cause the windmill to rotate, or does the windmill's rotation cause the wind? Physically, only one direction is sensible — the wind causes the windmill to rotate:

$$\text{wind} \rightarrow \text{windmill rotation}.$$

Here it is legitimate to regress the effect on the cause, with the dependent variable being the *speed of the windmill* and the explanatory variable being the *speed of the wind*:

$$\text{windmill}_i = \beta_0 + \beta_1 \text{wind}_i + u_i,$$

where u_i denotes other factors that may affect windmill speed (e.g. mechanical friction, maintenance). The slope $\beta_1 = \Delta \text{windmill} / \Delta \text{wind}$ is the causal effect of a one-unit change in wind speed on windmill speed, and regressing windmill speed on wind speed correctly identifies it.

Own example. Hours of study X_i and exam score Y_i : more hours of deliberate practice causally raise a student's score (holding ability fixed), $\text{score}_i = \beta_0 + \beta_1 \text{hours}_i + u_i$; regressing score on hours identifies β_1 provided no confound (e.g. innate ability) is left in u_i correlated with hours.

Case (ii): Y causes X (reverse causality). $Y \rightarrow X$: the arrow runs the opposite way from what the researcher intended to estimate. If the researcher nonetheless regresses Y on X , the "slope" obtained does not measure the effect of X on Y at all — it picks up (a scaled, noisy version of) the effect of Y on X .

Own example (the official slides give no example for reverse causality). Police staffing X_i and crime rate Y_i in a city-year panel: cities often raise police staffing *in response to* rising crime, i.e. crime $\rightarrow$ police, rather than police causing crime. A regression of crime

on police staffing can show a spurious *positive* correlation (more police, more crime) even though additional police plausibly *reduce* crime, because the causal arrow used by the policy process runs backward relative to the regression the researcher ran.

Windmill counter-check. If a researcher regressed *wind speed* on *windmill speed* instead, that would be the reverse-causality mistake for the same pair of variables — illustrating that the same correlated pair can be mis-specified as case (ii) if the direction is guessed wrongly.

Case (iii): a common factor C drives both X and Y (confounding). $C \rightarrow X$ and $C \rightarrow Y$ simultaneously, with no direct causal arrow between X and Y themselves. X and Y are correlated purely because they share a common cause C ; regressing Y on X directly is not meaningful, because there is no structural parameter “effect of X on Y ” to estimate — the correct object is the effect of C on each of X and Y separately.

Official example (ice cream and drowning). As ice cream sales increase, the rate of drowning deaths increases. Does eating ice cream cause drowning? No — there is no causal link. The common factor is *temperature*: hot weather independently raises both ice cream sales (people buy more ice cream when it is hot) and drowning deaths (more people go swimming when it is hot):

$$\text{temperature} \longrightarrow \text{ice cream sales}, \quad \text{temperature} \longrightarrow \text{drowning deaths}.$$

It does not make sense to build a single regression of ice-cream sales on drowning deaths (or vice versa). Instead the sensible models are two *separate* regressions, each on the confounder:

$$\text{icecream}_i = \beta_0 + \beta_1 \text{temp}_i + u_i, \quad \text{drowning}_i = \beta_0 + \beta_1 \text{temp}_i + u_i,$$

where icecream_i denotes ice-cream sales and drowning_i denotes the rate of drowning deaths (the two equations use generic parameter names β_0, β_1 but are two distinct regressions with two distinct sets of coefficients).

Official example (masks and online-working hours). The consumption of face masks and the hours spent working online are positively correlated. Neither variable causes the other; the common factor is COVID-19: the pandemic simultaneously drove up both mask purchases and remote/online working hours.

Own example. Across countries, the number of firefighters sent to a fire and the amount of damage caused by the fire are positively correlated. This does *not* mean firefighters cause damage; the common factor is *fire size* C — a bigger fire independently draws more firefighters *and* causes more damage, so $C \rightarrow X$ (firefighters) and $C \rightarrow Y$ (damage) with no direct arrow between X and Y .

Method 2: A single diagram distinguishing the three cases

The three cases can be summarised by where the arrowheads point in a three-node causal diagram with nodes X , Y , and (for case (iii)) an unobserved C :

$$\begin{array}{ccc} \underbrace{X \longrightarrow Y} & \underbrace{Y \longrightarrow X} & \underbrace{X \longleftarrow C \longrightarrow Y} \\ \text{(i) forward causality} & \text{(ii) reverse causality} & \text{(iii) common factor} \end{array}$$

This is not a new derivation but a compact restatement of Method 1 that is useful for quickly classifying a new empirical claim: ask “does a directed path exist from X to Y with no back-door path through an unobserved C ?” If yes, case (i); if the only directed path runs from Y to X , case (ii); if X and Y share a parent C with no direct arrow between them, case (iii).

What OLS of Y on X identifies in each case.

- **Case (i).** $\hat{\beta}_1$ from regressing Y on X consistently estimates the true causal slope β_1 , *provided* Least Squares Assumption 1 ($E[u | X] = 0$) holds, i.e. nothing else correlated with X is left in the error term.
- **Case (ii).** $\hat{\beta}_1$ from regressing Y on X does *not* estimate any causal effect of X on Y (there is none); at best it recovers information about the reverse effect of Y on X , contaminated by whatever generates X .
- **Case (iii).** $\hat{\beta}_1$ from regressing Y on X is generally *nonzero and statistically significant*, yet corresponds to no causal parameter at all — it is entirely an artefact of the shared dependence on C . Setting X by intervention (holding C fixed) would leave Y unchanged, which is exactly what a causal reading of $\hat{\beta}_1$ would get wrong.

Interpretation. Correlation is symmetric ($\text{Corr}(X, Y) = \text{Corr}(Y, X)$); causality is not. Before running $Y_i = \beta_0 + \beta_1 X_i + u_i$ and interpreting $\hat{\beta}_1$ causally, one must argue — using institutional knowledge, timing, or an explicit source of exogenous variation in X — that the data are generated by case (i) and not by case (ii) or (iii). This is precisely the role that Least Squares Assumption 1 ($E[u | X] = 0$) plays in later lectures: it is the formal condition under which regression-of- Y -on- X recovers a causal slope rather than a merely predictive one.

Beyond the slides. The official Tut 1 slides state the three cases with worked examples for case (i) (windmill) and case (iii) (ice cream/drowning; masks/online-working), but give no example for case (ii), do not draw the arrow diagrams above, and do not spell out what $\hat{\beta}_1$ from an OLS regression of Y on X would and would not identify in each case. This subsection (the police-staffing example, the itemised “What OLS identifies” list, the firefighter example, and the diagram in Method 2) fills that gap and previews the language (“ $E[u | X] = 0$ ”) that Lecture 3 will use to make the causality requirement precise.

3 cases when X, Y are correlated: (i) $X \rightarrow Y$, (ii) $Y \rightarrow X$, (iii) $C \rightarrow X, C \rightarrow Y$.

Question 2 (The class-size / test-score linear regression model)

Problem

Explain the linear regression model in the example of class size and student performance. What are the parameters of the model? How to interpret these parameters? What is the meaning of the error term? (*Tutorial Question Set No. 1, Q2.*)

Solution

Method 1: State the model and interpret each piece (official route)

Step 1: The model. Let TestScore_i denote the average standardised test score of school district i , and let STR_i denote district i 's student–teacher ratio (number of students divided by number of full-time-equivalent teachers). The simple linear regression model relating them is

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{STR}_i + u_i, \quad i = 1, \dots, n,$$

where $n = 420$ California school districts in the running example. TestScore_i is the *dependent* (explained) variable, STR_i is the *independent* (explanatory) variable / regressor, and u_i is the *error term*.

Step 2: The parameters. There are two parameters:

- β_0 , the **intercept**: the value of the population regression line $\beta_0 + \beta_1 \text{STR}$ at $\text{STR} = 0$. It need not have a direct real-world interpretation here, since no school district actually has a student–teacher ratio of zero; it simply anchors the line.
- β_1 , the **slope**, which is the parameter of central interest. It is the rate of change of the population regression line:

$$\beta_1 = \frac{\Delta \text{TestScore}}{\Delta \text{STR}},$$

i.e. the change in average test score associated with a one-unit (one-student-per-teacher) increase in the student–teacher ratio, holding the influence of u_i fixed.

Step 3: Interpreting the parameters — units and a numerical illustration. β_1 carries units of points of TestScore per unit of STR (points per student-per-teacher). Suppose, illustratively, $\beta_1 = -2$: then a $\Delta \text{STR} = -2$ change (two fewer students per teacher, e.g. from $\text{STR} = 20$ to $\text{STR} = 18$) is associated with a predicted test-score change of

$$\Delta \text{TestScore} = \beta_1 \times \Delta \text{STR} = (-2) \times (-2) = 4 \text{ points},$$

i.e. shrinking class size by 2 students per teacher raises the predicted average test score by 4 points. The *sign* of β_1 tells us the direction of the association (here, negative: larger classes go with lower scores), and the *magnitude* tells us how much test score moves per unit change in STR.

Empirical benchmark (Lecture 1–2 preview, not required for this conceptual question but useful context). The actual OLS fit on the $n = 420$ district dataset gives $\widehat{\text{TestScore}} = 698.9 - 2.28 \text{ STR}$, so the estimated $\hat{\beta}_1 = -2.28$ is close in spirit to the illustrative $\beta_1 = -2$ used above.

Step 4: The meaning of the error term. u_i represents *all other factors* besides the student–teacher ratio that affect district i ’s average test score: e.g. household income, parental education, school facilities, teacher quality, outside learning resources, and any other district-level heterogeneity not captured by STR_i . It is what makes the model *statistical* rather than a deterministic physical law — two districts with identical STR_i need not have identical TestScore_i , and the difference is absorbed into u_i .

Method 2: Conditional-expectation restatement

An equivalent way to state the same model, useful because it foreshadows Least Squares Assumption 1 (Lecture 3), is via the conditional population regression function. If $E[u_i | \text{STR}_i] = 0$, then

$$E[\text{TestScore}_i | \text{STR}_i] = \beta_0 + \beta_1 \text{STR}_i,$$

so $\beta_0 + \beta_1 \text{STR}$ is literally the *average* test score across all districts sharing a given student–teacher ratio, and β_1 is the change in that conditional average per unit change in STR . This route is not needed to answer Q2 as posed (which only asks to “explain” the model), but it makes explicit that β_1 is a statement about population averages, not about any individual district.

Interpretation. The model formalises the empirical claim “class size affects student performance” as a linear relationship between a population average of test scores and the student–teacher ratio, with β_1 as the object of policy interest (e.g. “would hiring more teachers to shrink class sizes raise test scores, and by how much?”). Whether $\hat{\beta}_1$, once estimated from data, can be read *causally* depends on whether u_i is uncorrelated with STR_i (Least Squares Assumption 1) — e.g. wealthier districts might both afford smaller classes *and* have other advantages (private tutoring, more engaged parents) baked into u_i , which would bias a naive reading of $\hat{\beta}_1$ as “the” causal effect of class size.

Beyond the slides. The official slides state the model and the roles of β_0, β_1, u_i but do not give a numerical illustration of β_1 with units, nor connect the model to the conditional-expectation form. Both are supplied here (Step 3 and Method 2) since the question explicitly asks “how to interpret these parameters,” which calls for more than a one-line description.

$\begin{aligned} \text{TestScore}_i &= \beta_0 + \beta_1 \text{STR}_i + u_i, & \beta_1 &= \Delta \text{TestScore} / \Delta \text{STR}, \\ u_i &= \text{all other factors affecting TestScore}_i. \end{aligned}$

Question 3 (Conditional and unconditional moments of $Y = X^2 + Z$)

Problem

Let X and Z be two independently distributed standard normal random variables, and let $Y = X^2 + Z$.

- (a) Show that $E[Y | X] = X^2$.
- (b) Show that $E[Y] = 1$.
- (c) Show that $E[XY] = 0$.

(Tutorial Question Set No. 1, Q3.)

Solution

Facts used throughout. Since $X \sim N(0, 1)$: $E[X] = 0$; $\text{Var}(X) = E[X^2] - (E[X])^2 = 1 \Rightarrow E[X^2] = 1$; the third central moment (and, since the mean is 0, the third raw moment) vanishes by symmetry of the normal density, $E[X^3] = 0$ (odd moments of a symmetric-about-0 distribution are zero); the fourth moment is $E[X^4] = 3$ (kurtosis of a standard normal), not needed below but recorded for completeness. Likewise $Z \sim N(0, 1)$ gives $E[Z] = 0$. Because X and Z are *independent*:

$$E[g(X)h(Z)] = E[g(X)]E[h(Z)] \quad \text{for any functions } g, h,$$

and, for conditioning, independence gives $E[Z | X] = E[Z] = 0$ (knowing X gives no information about Z).

Method 1: Direct computation (official route)

Part (a): $E[Y | X] = X^2$. Condition on X and use linearity of conditional expectation:

$$E[Y | X] = E[X^2 + Z | X] = E[X^2 | X] + E[Z | X].$$

Given X , the term X^2 is a known (non-random) function of the conditioning variable, so $E[X^2 | X] = X^2$. By independence of X and Z , $E[Z | X] = E[Z] = 0$. Hence

$$E[Y | X] = X^2 + 0 = X^2.$$

$E[Y | X] = X^2.$

Part (b): $E[Y] = 1$. Using linearity of (unconditional) expectation directly on $Y = X^2 + Z$:

$$E[Y] = E[X^2 + Z] = E[X^2] + E[Z] = 1 + 0 = 1.$$

$E[Y] = 1.$

Part (c): $E[XY] = 0$. Substitute $Y = X^2 + Z$ and expand:

$$E[XY] = E[X(X^2 + Z)] = E[X^3 + XZ] = E[X^3] + E[XZ].$$

The first term is $E[X^3] = 0$ by the odd-moment fact above. For the second term, independence of X and Z gives $E[XZ] = E[X]E[Z] = 0 \times 0 = 0$. Hence

$$E[XY] = 0 + 0 = 0.$$

$E[XY] = 0.$

Method 2: Via the Law of Iterated Expectations (LIE)

This route is genuinely different from Method 1: instead of computing each unconditional moment from scratch, it *reuses* the conditional result from part (a) and pushes it through the Law of Iterated Expectations, $E[W] = E[E[W | X]]$ for any random variable W and conditioning variable X .

Part (b) via LIE.

$$E[Y] = E[E[Y | X]] = E[X^2] = 1,$$

using part (a)'s result $E[Y | X] = X^2$ in the first equality and $E[X^2] = 1$ (standard normal second moment) in the second. This reproduces $E[Y] = 1$ without ever expanding $Y = X^2 + Z$ again.

Part (c) via LIE. Apply the LIE to the product XY , conditioning on X and using that X itself is a known constant once we condition on X (so it can be pulled out of the inner conditional expectation):

$$E[XY] = E[E[XY | X]] = E[X E[Y | X]] = E[X \cdot X^2] = E[X^3] = 0,$$

again reusing part (a)'s $E[Y | X] = X^2$, and then the odd-moment fact $E[X^3] = 0$.

Interpretation. Method 2 shows that once the conditional mean $E[Y | X] = X^2$ is established, both unconditional moments in parts (b) and (c) follow immediately by "integrating out" X against the conditional expectation — there is no need to re-derive them from the definition of Y each time. This is the general LIE pattern $E[W] = E[E[W | X]]$ and $E[XW] = E[X E[W | X]]$ that recurs throughout the course (e.g. in deriving that Least Squares Assumption 1, $E[u | X] = 0$, implies $\text{Corr}(u, X) = 0$).

Beyond the slides. The official Tut 1 slides present only Method 1 (direct substitution), compressed into a few lines with each fact ($E[X] = 0$, $E[X^2] = 1$, $E[X^3] = 0$, independence) stated but not re-derived. This solution (i) makes every step's justification explicit (which moment fact, which independence property, at which line); (ii) adds Method 2 via the LIE as a structurally different, reusable route; and (iii) adds the following observation, which the slides do not make at all.

Notice that $\text{Corr}(X, Y)$ is computed from $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - 0 \times 1 = 0$, so X and Y are **uncorrelated** ($\text{Corr}(X, Y) = 0$) — and yet $Y = X^2 + Z$ depends *very strongly* on X (indeed $E[Y | X] = X^2$ is a deterministic, highly nonlinear function of X : knowing X pins down half of Y 's value exactly). This is a clean illustration that **zero correlation does not imply independence**, and more specifically that zero correlation does not imply $E[Y | X]$ is constant in X . Linear correlation is blind to this

quadratic (even, symmetric) dependence, since the positive dependence for $X > 0$ and $X < 0$ regions cancel out in the linear covariance sum.

This has direct relevance to **Least Squares Assumption 1** introduced in Lecture 3: the assumption required for OLS to recover a causal/structural slope is the much stronger condition $E[u | X] = 0$ (zero *conditional mean* of the error, for *every* value of X), not merely $\text{Cov}(u, X) = 0$ (zero correlation). This question is a worked counterexample showing the gap between the two: here Z plays the role of a well-behaved error term relative to X ($E[Z | X] = 0$ holds), but if one instead tried to treat X^2 itself as an "error" relative to some other regressor, zero correlation alone would badly understate how much information that regressor carries about Y .

$E[Y X] = X^2, \quad E[Y] = 1, \quad E[XY] = 0,$ yet $\text{Corr}(X, Y) = 0$ while Y depends strongly on X.

Question 4 (Type I and Type II errors and the trade-off)

Problem

Explain the Type I and Type II errors and the trade-off between the two errors by using a specific example. (*Tutorial Question Set No. 1, Q4.*)

Solution

Method 1: Definitions, 2×2 table, and the COVID ART-test example (official route)

Step 1: Definitions. Let H_0 denote the null hypothesis and H_1 the alternative hypothesis. A hypothesis test is a decision rule that, based on sample data, either rejects or does not reject H_0 . Two kinds of mistakes are possible:

- **Type I error** (false positive): reject H_0 when H_0 is in fact true.
- **Type II error** (false negative): fail to reject H_0 when H_1 is in fact true.

The **significance level** of a test is $\alpha = P(\text{Type I error}) = P(\text{reject } H_0 \mid H_0 \text{ true})$; common choices in economics are 10%, 5%, or 1%.

Step 2: The 2×2 decision table.

	H_0 true	H_1 true
Do not reject H_0	Correct (prob. $1 - \alpha$)	Type II error (prob. β)
Reject H_0	Type I error (prob. α)	Correct / “power” (prob. $1 - \beta$)

Here α is the significance level and $1 - \beta$ is the *power* of the test (probability of correctly rejecting H_0 when H_1 is true).

Step 3: The COVID-19 ART (antibody rapid test) example. Let

$$H_0 : \text{healthy (no COVID)}, \quad H_1 : \text{sick (has COVID)}.$$

The test measures the antibody level in a person’s body and compares it against a **threshold** (critical value): the test *rejects* H_0 (declares the person infected) if the measured antibody level exceeds the threshold. Suppose the original threshold is 10 units, i.e. the test rejects H_0 whenever the antibody level is above 10.

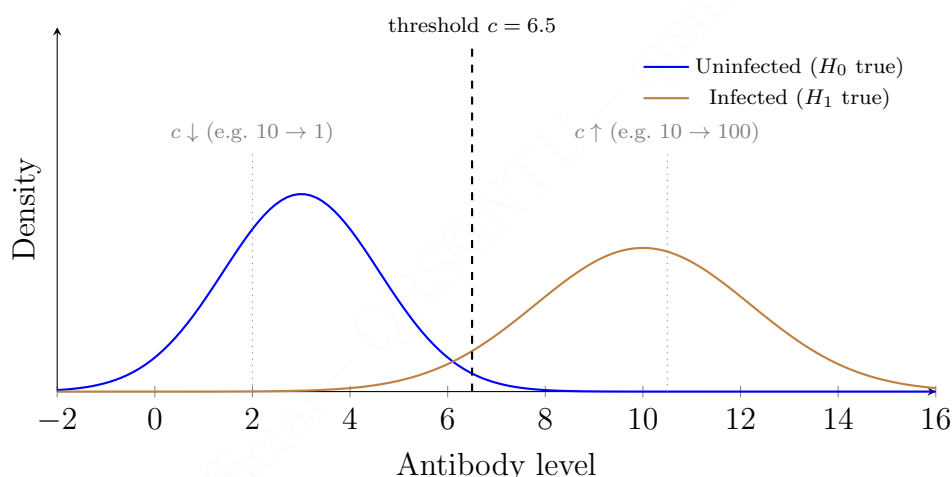
Raising the threshold from 10 to 100. It now becomes much harder to reject H_0 (only very high antibody levels, above 100, trigger a positive result). This **decreases the Type I error**: fewer healthy people are wrongly flagged as infected. But it simultaneously **increases the Type II error**: many genuinely infected people, whose antibody levels lie between 10 and 100, are now missed (false negatives) — the test loses detecting power.

Lowering the threshold from 10 to 1. It now becomes very easy to reject H_0 (any antibody level above just 1 unit triggers a positive result). This **decreases the Type II error**: the test becomes very sensitive and catches almost all truly infected people. But it simultaneously **increases the Type I error**: many healthy people, whose natural antibody levels lie between 1 and 10, are now wrongly flagged as infected (false positives).

Step 4: The trade-off. In general we cannot simultaneously minimise both error probabilities using only the choice of threshold: moving the threshold to the right (stricter, larger critical value) trades a lower Type I error for a higher Type II error, and moving it to the left (looser, smaller critical value) trades a lower Type II error for a higher Type I error. This is exactly the trade-off visualised by the two overlapping antibody-level distributions in the official figure (redrawn in Method 2 below): the uninfected population's antibody levels are typically low, the infected population's are typically high, but the two densities *overlap*, so any single threshold necessarily leaves some overlap area on the "wrong side" for both groups.

Method 2: Redraw the official figure — overlapping antibody-level densities with a movable threshold

The official Tut 1 answer to this question *is* a figure (two overlapping densities with a threshold line) rather than additional text; we redraw it in pgfplots so the trade-off is visible directly, together with a second, movable threshold to show both directions of Step 3.



The area under the blue (uninfected) curve to the *right* of the threshold c is the Type I error probability α (healthy people wrongly flagged positive); the area under the brown (infected) curve to the *left* of c is the Type II error probability β (infected people wrongly flagged negative). Moving c right (toward $c \uparrow$) shrinks the blue tail past c (lower α) but grows the brown area left of c (higher β); moving c left (toward $c \downarrow$) does the reverse. Because the two densities overlap over a whole range of antibody levels, no single choice of c can drive both tail areas to zero simultaneously — this overlap is the geometric source of the Type I/Type II trade-off.

Step 5: A second example — court verdicts. Let H_0 : the defendant is innocent, H_1 : the defendant is guilty. "Reject H_0 " corresponds to a guilty verdict.

- **Type I error:** convicting an innocent defendant (false positive) — widely regarded as the more costly error in criminal law.
- **Type II error:** acquitting a guilty defendant (false negative).

The "threshold" here is the standard of proof. Common-law systems deliberately set a very demanding standard ("beyond reasonable doubt") for criminal cases, which pushes

the threshold toward making Type I errors rare (few innocents convicted) at the cost of more Type II errors (some guilty defendants acquitted for lack of sufficiently strong evidence) — the same structural trade-off as the ART test, just with the threshold set by legal doctrine rather than a numeric antibody cut-off. Civil cases use a lower standard (“preponderance of the evidence”), shifting the trade-off in the opposite direction.

Interpretation. Both examples show that the choice of significance level α (equivalently, the choice of threshold/standard of proof) is a policy decision, not a purely statistical one: it encodes how the costs of the two error types are weighed against each other. A criminal court accepts more false acquittals (Type II errors) to keep false convictions (Type I errors) rare, because wrongly imprisoning an innocent person is judged far costlier than letting a guilty person go free.

Beyond the slides. The official Tut 1 slides state the trade-off using the $10 \rightarrow 100$ and $10 \rightarrow 1$ threshold-shift narrative for the ART test and show a figure, but the figure itself is not text-recoverable from the source PDF (see `HE2003_digest.md`, §4, Tut 1). This solution (i) redraws the density figure explicitly with a movable threshold in Method 2, labelling the Type I and Type II error regions as tail areas; (ii) adds the court-verdict example as a second, non-numeric illustration of the same trade-off; and (iii) connects both examples explicitly to the economics convention of choosing $\alpha \in \{10\%, 5\%, 1\%\}$: just as courts pick a standard of proof to control the Type I-error rate, econometricians conventionally fix $\alpha = 5\%$ (sometimes 1% for stronger evidence requirements) when testing hypotheses such as $H_0 : \beta_1 = 0$ in later lectures, accepting a correspondingly higher Type II error (lower power) in exchange for controlling false rejections of a true null.

$\begin{aligned}\text{Type I error} &= P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha, \\ \text{Type II error} &= P(\text{fail to reject } H_0 \mid H_1 \text{ true}) = \beta, \\ \alpha \downarrow &\Rightarrow \beta \uparrow \text{ (and vice versa).}\end{aligned}$

Method Map

Q.	Core topic	Methods included
1	Correlation vs. causality	Enumerate the 3 structural cases with examples (Method 1); compact arrow-diagram restatement and "what OLS identifies" checklist (Method 2).
2	Class-size regression model	State model, parameters, units, numerical illustration (Method 1); conditional-expectation restatement (Method 2).
3	Moments of $Y = X^2 + Z$	Direct computation via independence and normal moments (Method 1); Law of Iterated Expectations reusing part (a) (Method 2).
4	Type I/II errors and trade-off	Definitions, 2×2 table, COVID ART-test threshold shifts (Method 1); redrawn overlapping-density figure with movable threshold, plus court-verdict example (Method 2).

Technique Summary

- Classify before you regress.** Before running $Y_i = \beta_0 + \beta_1 X_i + u_i$ and reading $\hat{\beta}_1$ causally, identify whether the correlation between X and Y arises from forward causality, reverse causality, or a common confounder C — only the first case licenses a causal reading, and even then only under Least Squares Assumption 1.
- A regression model has three named pieces, always.** Dependent variable, regressor(s), and error term; the slope parameter answers "how much does the dependent variable change per unit change in the regressor," carrying whatever units that ratio implies.
- The Law of Iterated Expectations turns conditional results into unconditional ones for free.** Once $E[Y | X]$ is known, both $E[Y] = E[E[Y | X]]$ and $E[XY] = E[X E[Y | X]]$ follow without re-deriving from the definition of Y .
- Zero correlation is weaker than zero conditional mean.** $\text{Cov}(X, Y) = 0$ can coexist with Y depending deterministically on X (as in $Y = X^2 + Z$); the OLS causality requirement $E[u | X] = 0$ is the stronger, correct condition.
- Type I and Type II errors trade off through the threshold/critical value.** Moving the decision threshold to reduce one error type mechanically increases the other whenever the two underlying distributions overlap; the choice of significance level α is where this trade-off is made explicit.

Final Answers

- Three cases when X, Y are correlated: (i) $X \rightarrow Y$ (e.g. wind $\rightarrow$ windmill rotation); (ii) $Y \rightarrow X$ (reverse causality, e.g. crime $\rightarrow$ police staffing); (iii) common factor $C \rightarrow X, C \rightarrow Y$ (e.g. temperature $\rightarrow$ {ice-cream sales, drowning deaths}; COVID $\rightarrow$ {mask use, online-working hours}).

2. $\text{TestScore}_i = \beta_0 + \beta_1 \text{STR}_i + u_i$; β_0 is the intercept (value at $\text{STR} = 0$); $\beta_1 = \Delta \text{TestScore} / \Delta \text{STR}$ (points per unit STR; e.g. $\beta_1 = -2 \Rightarrow \Delta \text{TestScore} = 4$ for $\Delta \text{STR} = -2$); u_i is all other factors affecting test score.
3. $E[Y | X] = X^2$; $E[Y] = 1$; $E[XY] = 0$; yet $\text{Corr}(X, Y) = 0$ while Y depends strongly (nonlinearly) on X .
4. Type I error = $P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$; Type II error = $P(\text{fail to reject } H_0 \mid H_1 \text{ true}) = \beta$. Raising the ART threshold ($10 \rightarrow 100$) lowers α but raises β ; lowering it ($10 \rightarrow 1$) reverses this — court verdicts show the same trade-off via the standard of proof.